Housing First or Treatment First? Ask the Kids, Not the Parents

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Letter to the Coordinating Editor

Summary of changes

Dear Professor Karabarbounis,

Thank you for your decision letter, and for leaving the door open to an almost new paper. That is what I am resubmitting. The model, the calibration, the validation and the experiments have all been rebuilt, and the paper has a new title. This letter summarises what changed and which comments each change answers. Four further letters respond to the Associate Editor and to each referee comment by comment, with the responses in red. Section, table and figure numbers refer to the new paper. Comments are numbered as in those letters: A1 to A20 for the Associate Editor, and R1, R2 and R3 for the referees.

What the paper now finds

The paper is now *Housing First or Treatment First? Ask the Kids, Not the Parents*, and the title is its answer. The version you rejected found that, at the same generosity, Treatment First reduced the disorder but raised homelessness, and it ranked universal Treatment First above universal Housing First only once a loss of consortium adjustment was added to welfare after the model was solved. The new paper drops the loss of consortium and asks whether moral hazard, the argument that dominates the policy debate, justifies making housing assistance conditional on sobriety.

It does not. Without any effect on children, a newborn is worse off under Treatment First by 1.09 percent of lifetime consumption, even though drug use falls and the drug test's savings buy vouchers for other households. Conditionality takes an option away from forward looking households who weigh the risk of a substance use disorder (SUD) correctly, and gives them back less through the government's budget than it takes.

The answer turns instead on the children, through three externalities that a parent counts only in part. A child shares its parent's consumption and dwelling but not the drug, so what the parent spends on drugs, and the earnings it loses while a SUD keeps it out of work, the child goes without: the budget externality. A child whose parent had a SUD while it lived at home starts adult life with lower productivity: the earnings externality. And a child lives wherever its parent lives, so a revoked voucher can put it in a shelter: the childhood homelessness externality, the one critics of Treatment First expect it to create. Within the existing rationed programme, with the drug price and the probability of drawing a voucher both clearing, Treatment First cuts drug use by 21 percent and the share of households with a SUD by 19 percent, and it lowers homelessness and the share of children in a shelter by 11 percent each. A newborn gains 2.57 percent of lifetime consumption, three quarters of it an externality its parent does not count. The parents gain too, by 0.69 percent of their own lifetime consumption, because the higher chance of drawing a voucher and the cheaper drug outweigh the option the screen removes. Along the transition, neither homelessness nor the share of children in a shelter rises in any year, and children born at the reform gain 1.45 percent.

The model reproduces the micro evidence. Without being targeted, it comes close to the effect of making housing conditional on abstinence in the one trial that compares a sobriety condition with unconditional housing (1.40 against 1.26; Milby et al., 2005), and in the model losing a voucher raises the three year mortality of households with serious mental illness (SMI) by about as much as Housing First lowered it among homeless veterans with mental illness (3.6 against 4.6 percentage points; Costantini, 2025). As in that evidence, households with a SUD lose: in the comparison a trial would make, with the drug price and the voucher probability held, 74 percent of those with a SUD when the reform lands are worse off. What reverses the verdict is what trials do not observe: the households the condition deters from ever starting, the vouchers its savings buy for others, and the children.

The channel that decides the comparison came from the reports. Referee 2 (C3) asked for parents who care about their children's prospects, and expected it to amplify the gain from reducing the disorder. The Associate Editor (A6) objected that the previous version discussed intergenerational transmission with no link between generations in the model. Built, the channel does more than amplify: it reverses the ranking.

What was built

A stylized model (Section 3), with two periods and calibrated (Tables 1 and 2), shows why drug use concentrates at the bottom of the income distribution when no preference varies with income (A1). Drugs are a luxury good, so without consequences the average user would earn 3.51 times the population average. The risk of dying that a SUD brings cannot be insured, and its cost grows with income; on its own it takes that figure to 0.64 at its calibrated strength. The risk of unemployment, which a household can partly insure by saving, leaves it at 3.51 even at twice its strength (Figure 1). Referee 1 asked for this comparison in the full model (1(d)). I make it in a model with saving and with consequences that arrive later, the two things the old static appendix lacked (1(e)), and in the quantitative model the mortality multiplier of a SUD is calibrated to the income of users instead (Section 5.1).

A quantitative model (Section 4), life cycle and annual, in general equilibrium in the drug price and in the government's budget. Three things in it carry the paper.

- *Children.* A child lives with its parent for 25 years, shares its consumption and dwelling, and enters the parent's flow utility at the weight ω_c. A SUD the parent has while the child lives at home lowers the child's productivity for life, and the parent is charged for that loss as a difference against a child whose parent never had the disorder (Sections 4.2, 4.8 and 4.10).
- *Preferences that are the same in every state.* No preference parameter varies with the disorder, with SMI or with housing (Section 4.2), so reducing the disorder or housing more people raises no one's utility mechanically, which was the objection in Referee 2's C5, Referee 3's M2(a) and the Associate Editor's A8. The disorder works through what it costs: mortality, employment, recovery and the child's productivity. A child inherits a worse productivity draw, not a taste for drugs.
- *Recovery that depends on housing.* Referee 2 (C1), the Associate Editor (A13) and Referee 1 (2(d)) asked for discipline on the response of drug use to housing access. In the model it runs through recovery, which is less likely in a shelter by a factor calibrated to Milby et al.'s (2005) finding that housing raises the chance of recovering by 47.3 percent (the model gives a ratio of 1.45 against 1.47). It is a transition probability rather than a taste, and it is what the trial

measures.

What was cut, and what was added

The Associate Editor asked whether a list of ingredients earned their place. I cut or simplified each of them but one, the link between generations, which the paper now models and which carries the result. Earlier versions solved the model with endogenous labour supply, rents, homeownership and house prices, and these margins had a negligible effect on the results (footnote 19).

Ingredient	Now	Comments
Homeownership, mortgages and house prices	Cut. Households rent. The grid is the shelter and three rental sizes; the shelter and the smallest rental are calibrated, and the two larger sizes are set at four and eight times the smallest (Table A1).	A4; R2 D1
Endogenous rents	Cut. Equilibrium is now where the Associate Editor asked for it, in the government's budget and the drug price, and Table 5 switches each on and off.	A5
Labour supply	Cut. Labour is supplied inelastically when employed.	A3; R3 m1
Unemployment insurance	One year of benefits for the share of job losers who are insured, nothing after (Section 4.3).	A2
Type dependent transfers and bequests	One endowment at entry, the same for every entrant, from the estates of the dead; the government collects the rest.	A7
Fiscal costs of homelessness and of a SUD	Cut. The shelter is free, and the budget counts what the government pays in transfers, drug tests and treatment (Section 4.11).	A15; R2 C6
Light and heavy use	Cut. One chance that a year of use leads to a SUD (footnote 20).	
Loss of consortium	Cut (below).	A9; R1 3(a) to 3(c)
Universal vouchers	Cut. Every experiment keeps the programme rationed, with the voucher probability adjusting to hold net revenue at its benchmark (below).	R2 C6; R3 m3

Against that, the model gained what the reports asked for.

1. **SMI** (R1 1(a); Section 4.7): a persistent state whose onset a SUD makes twice as likely, and which lowers the chance of finding a job, raises mortality and slows recovery from a SUD. Without being targeted, households with SMI are 2.64 times as likely as others to use drugs, against 2.44 in the data. Referee 1 expected it to weaken Treatment First; the screen does fall hardest on households with SMI, and Treatment First still raises welfare.
2. **Treatment** (R1 1(b); A10; Section 4.6): a choice, at a utility cost calibrated to a take up of 24.5 percent, that doubles the chance of recovering and that the government pays for. Under Treatment First a household with a SUD passes the screen only in treatment and without using, and the share in treatment rises from 24.6 to 29.0 percent. If a household with a SUD cannot pass the screen at all, Treatment First still lowers homelessness by 11 percent and a newborn gains at least as much (footnote 38).
3. **An equilibrium drug price** (R1 1(c); R3 M2(d); A5; Section 4.12), clearing a supply curve with an elasticity of 1.46 (Mace et al., 2020). It works against Treatment First: the price falls

by 5 percent and draws users back, which takes the newborn's gain from 6.45 to 2.57 percent.

4. **Imperfect and costly enforcement** (R1 4(a) and 4(c); A11; Sections 4.9 and 6). The screen catches a user with probability 20 percent a year, set so that it deters about as much as the US military's random testing did: 62 percent of the users facing it stop, against a fall of 61 percent in past year marijuana use (Mehay and Pacula, 1999). A test costs 35 dollars, the upper end of the estimates, and a household that is caught is excluded for three years on average. At half the detection rate, Treatment First still cuts drug use by 14 percent and homelessness by 6 percent, and a newborn gains 1.65 percent.
5. **Transitions** (R1 3(d); Sections 6.4 and 6.5). The transition starts from the whole Housing First population alive at the reform and clears the drug price and the voucher budget every period (Figure 3). Drug use falls on impact, 81 percent of its long run fall in the first year, and homelessness and the share of children in a shelter never rise; 11 percent of the adults alive at the reform are worse off. When a household with a SUD cannot stop using (Table 7), the design matters: a screen that catches every user from the first day raises homelessness by up to 14 percent, while at the calibrated detection rate, exempting households in treatment for the first five periods holds the rise to 1.6 percent and keeps the share of children in a shelter from rising at all. Children born at the reform gain in every case.
6. **The disamenity of homelessness** (R2 C2). It is not in the benchmark. With one at Loubière et al.'s (2020) contingent valuation, about 0.2 percent of average earnings a year, no long run gain in Table 5 moves by more than 0.14 percentage points (footnote 4).
7. **The productivity cost of a SUD** (A20). The disorder lowers a user's earnings by keeping it out of work longer (Section 4.8); there is no separate wage penalty on the user. The productivity effect the model carries is on the child, set to Nilsson's (2017) natural experiment estimate, 0.24. Even at zero, Treatment First raises welfare (footnote 5).

The loss of consortium adjustment is gone

Referee 1 (3(a) to 3(c)), Referee 3 (M2(c)) and the Associate Editor (A9) all objected to it: an adjustment applied after the model was solved, with no moment behind it, which changed the ranking, and whose conservative value is arguably zero. I agree, and it is deleted. Welfare is now a newborn's own expected lifetime utility, childhood included (Section 6.1), and nothing is added to it after the model is solved.

What decides the ranking now lives inside the household's problem. A parent counts the share ω_c of what its choices cost its child and acts on it; the rest is the externality. The weight is calibrated to behaviour: in the 2024 National Survey on Drug Use and Health (NSDUH), adults living with one other adult and no child are 17.5 percent more likely to use drugs than comparable lone parents, and ω_c is the weight at which removing a parent's concern for its child raises drug use in the model by as much (Section 5.1, Block 5). At Nishiyama's (2000) weight of 0.69, the upper end of the literature, a newborn gains more, 3.18 percent, though less of it is an externality (footnote 28). Section 6.3 then does what Referee 3 asked in M2(c): it splits the newborn's gain into the childhood externality, 1.50 percent of lifetime consumption, the earnings externality, 0.46, the newborn's own adult life under the policy, 0.43, and the part its parent counts, 0.19 (Table 6).

On tone (Referee 1, 3(c)), the paper says who loses as plainly as who gains, and its conclusion is conditional: yes if the externalities on children are counted, no if the case rests on moral hazard

(Section 7). In the long run 74 percent of households are better off and 3 percent worse off, but only 24 percent of those with a SUD are better off (Section 6.2).

[Still open.] The Associate Editor suggested reporting welfare and mortality separately, as a way round the adjustment. With the adjustment gone, mortality enters welfare only through the value households place on their own lives, which is calibrated to the value of a statistical life. The paper checks the model's mortality response against Costantini (below), but it does not report mortality as an outcome of the experiments.

Errors in the previous version

[Conceded.] Rebuilding the calibration from the data turned up errors in the version you rejected. These are the ones that bore on the results.

1. **The homelessness target was a rate of people**, while the model's agent is a household. The paper now targets a band of 0.40 to 1.00 percent of households, from the Department of Housing and Urban Development's single night count to the share homeless at some point in a year (Burt et al., 2001), and the model gives 0.62 percent.
2. **The share of the homeless with a SUD**, the 37 percent the Associate Editor questioned (A16), was the midpoint of a wide range of substance abuse rates, not a rate of drug use disorder. The paper now targets 21.7 percent, the drug use disorder share of the homeless in Gutwinski et al. (2021), and the model gives 20.2. Homelessness means literal homelessness, a shelter or the street (Section 4.5).
3. **The unemployment rate given a SUD was too high.** In the 2024 NSDUH it is 16.2 percent for adults aged 26 and over. It is no longer a target, and the model gives 14.5 percent (Table 4).
4. **Drug use, the income of users and drug spending came from different definitions of a user.** A user is now an adult aged 26 or over whose use amounts to a disorder of any severity, 8.90 percent in the 2024 NSDUH. The income of users and the share of users with a SUD are measured on that population, and drug spending is taken over its income (Section 5.1, Block 1).

Calibration and validation

Referee 1 could not assess the quantification because the calibration table never reported the model's moments (2(a), 2(b)), and the Associate Editor agreed (A12). Of the model's 59 parameters, the 26 that carry the story are in Table 3, in six blocks, and each entry says what the parameter controls, how it bears on the two policies and what happens as it rises, before its value, moment and source (A17); the hazards of death and unemployment are set out the same way, each with its source or moment (R3 M1(b)). The 33 standard ones are in Appendix A (Table A1). Both tables report the data, the model and the value for every target. The replication package documents the dataset, year, universe and computation behind every estimate, and carries the data (A17; R1 2(e)).

Most targets are close. The largest gaps are the income of users, 62 percent of the average in the model against 74 in the data, the share of renters paying more than half their income, 25.8 against

19.2 percent, and the unemployment rate, 5.6 against 4.9 percent. **[Conceded.]** The income of users is the gap Referee 1 raised in 2(c)(i), where the previous model put it at 41 percent. He was right that the model made users too poor, and it still does, by less. A new figure (Figure 2) sets the share using drugs at each level of income against the 2024 NSDUH: use is highest at the bottom and lowest at the top, as in the data, but in the model it is higher than the data at the bottom and flatter in the middle, and both lower the income of users (R3 M1(c) and M2(b)).

The policy responses the reports asked the model to match (R1 2(d); R2 C1; A13) are now in the paper. Two are calibrated: the response of recovery to housing, to Milby et al. (Table 3), and the response of drug use to a drug test, to the military's testing (Section 6). Two are not, and the new validation section (Section 5.2, Table 4), which also compares the homeless, users and voucher holders in the model with the data (R3 M1(a)), shows the model comes close to both: Milby et al.'s effect of making housing conditional on abstinence (1.40 against 1.26) and Costantini's mortality effect (3.6 against 4.6). The price elasticities are no longer targets (R1 2(e); R3 M2(d)): participation responds more than in the evidence, -1.32 against -0.3 to -0.9, which works against Treatment First, since the price it lowers draws users back. Transmission across generations is at the low end of the evidence for a SUD, an odds ratio of 1.46 against 1.44 to 2.51, and above it for homelessness, 6.23 against 1.8 to 2.4, so the model loads the effect on children towards the margin Housing First acts on.

The misses are reported as misses (R1 2(b), 2(c)). The homeless in the model are poorer and younger than in the data: 21.4 percent are employed against 46, their income is 6.5 percent of the average against 19.1, and 11.5 percent are 55 or older against 25.9. The paper reports these as gaps and says what each means for the experiments.

Costantini (2025)

The Associate Editor (A14) read Costantini as finding that Housing First lowers the three year mortality of homeless veterans by 4.6 percentage points while Treatment First has no lasting effect on health, in contrast with the model. Validation Block 7 now asks the model his Housing First question. With the drug price and the voucher probability held, losing a voucher raises the three year mortality of the households with SMI whom Treatment First's revocation puts on the street by about 3.6 percentage points, against his 4.6, without being targeted. The paper reports only this arm. His Treatment First arm is the Department of Veterans Affairs' own programmes, which the model does not represent, since its Treatment First is a voucher whose retention depends on sobriety (Section 4.9), and I do not claim a counterpart to it.

So the model agrees with him that losing housing raises mortality for the households who lose it. What his design does not observe, and the model does, is everyone else: in the long run Treatment First lowers homelessness by 11 percent, so fewer households are on the street at all.

Where I push back, and what the paper does not do

Costantini is the first place, above. The others:

1. **Labour supply.** The Associate Editor (A3) asked what it was doing in the model, and Referee 3 (m1) wanted it to price the cost of balancing the budget with labour taxes. I took the first view. Taxes stay at their benchmark values in every experiment and the voucher

probability balances the budget, so no labour tax moves, and in earlier versions endogenous labour supply had a negligible effect.

2. **The age limit on starting to use** (R1 S1). It stays, at age 80, and stops only households without a SUD from starting. Without it, households close to the model's last age, who know how little life remains, use too much (footnote 21).
3. **[Still open.] Generosity.** Referee 2 (C6) asked about optimal voucher design, Referee 3 (m3) for a continuous experiment in the voucher probability, and the Associate Editor found the rationed and universal comparisons the most relevant. The paper now asks one question, conditionality, within the existing rationed programme and at the same net cost. It does not study how generous the programme should be.
4. **[Still open.] False positives** (R1 4(b)). The model has none, because a positive screen is confirmed in a laboratory before it has consequences (Section 6), and there is no sensitivity to them. Stigma and administrative burden (4(c)) are not modelled beyond the cost of a test, set at the upper end of the estimates.
5. **[Still open.] Fiscal costs** (A15; R2 C6). Rather than defend the fiscal cost of homelessness that both found high, I removed it, so the paper makes no comparison with Abramson and Van Nieuwerburgh.
6. **[Still open.] Identification and sensitivity.** The paper reports neither the numerical identification Referee 1 asked for (2(f)) nor a systematic sensitivity analysis (4(d)). It reports the runs that bear on the ranking: half the detection rate, the parent's weight at the upper end of the literature, no productivity penalty for the child, households with a SUD unable to pass the screen, and a disamenity of homelessness. Treatment First raises a newborn's welfare in each.

Presentation

The body carries one comparison, Treatment First against Housing First in the rationed programme, taken apart step by step in Table 5, decomposed in Table 6 and followed along its transition in Figure 3 and Table 7. No experiments sit in an appendix, which holds only the standard parameters (A18). The word "addicted" is gone (A19). Table 5 reports every outcome in levels under each policy, and the text gives the changes in percent (R1 S2). The typographical errors Referee 1 listed (S3 to S10) were in passages and references that have been rewritten or removed. **[Still open.]** There are no plots of the drug use policy function (R2 C4) and no sensitivity graphs (R3 m2); the paper shows who uses, and why, through Figures 1 and 2.

Thank you again for the second look.

Sincerely,

Shahar Rotberg